



INTRODUCTION TO CARBONET

Cleaner Water Treatment Chemistry

CarboNet ●●

**Water treatment was a line item.
Now it's a liability.**

Your plant’s chemistry was built for steady-state conditions that no longer exist.

 American Water Works Association



State of the Water Industry report shows persistent concerns about funding

 SMARTCITIES DIVE

Aging infrastructure, climate risks shake water sector confidence: report

Century-old systems, extreme weather, cyberthreats and regulatory

The Washington Post
Democracy Dies in Darkness

ly and resilience, the American Water

The world has entered a new era of ‘water bankruptcy,’ U.N. report says

Researchers say this is not a temporary crisis but a permanent failure that requires rethinking the world’s approach to water scarcity.

Your plant's chemistry was built for steady-state conditions that no longer exist.

Water and wastewater treatment is no longer a managed, predictable function. Chemistry supply was an operational cost you contained—not a strategic exposure you managed.

That world is over.

- Tighter **discharge regulations** at every level of government
- Accelerating water scarcity emerging in new regions
- **Aging infrastructure** well past its design life
- **No capital budgets** to match
- Older equipment, smaller crews
- Plants are being asked to hit **stricter KPI**

Water treatment has stopped being an afterthought. It's become the thing operators lose sleep over.

Something has to give.

For facilities where it already has, operators don't run the process—**they are the process.**

The cost of the mismatch is compounding:

- More polymer to compensate for inconsistent performance
- More labor chasing settings and cleaning up after drift
- More sludge volume, more hauling, more compliance risk

Facilities pulling ahead have made a different trade:

- Chemistry that adapts to pH swings, variable solids, and flow changes
- No constant recalibration—performance holds as conditions shift
- Operators run the process. They don't chase it.

Only when your plant hits spec, are operators free from babysitting it.

When chemistry performs consistently, everything downstream changes:

- Crews show up to run the process—not manage make-down, chase pump settings, or clean up emulsion drift
- Presses hit dryness targets even as influent varies
- KPIs stay in spec without manual intervention
- Hauling volume drops; cost-per-ton treated falls
- Regulatory exposure falls alongside chemical load
- When an auditor asks how your program handles variability, you have a defensible, data-backed answer

The barriers between today and that future aren't operational.

They're structural.

Make-down equipment. Limits throughput, breaks down, ties up crews. Without it, most chemistry can't be applied at all.

Commodity PAM. Built for steady-state. Can't adapt to variable pH, solids spikes, or temperature swings—so operators compensate manually, all day.

Aging equipment. CAPEX isn't coming. Chemistry has to hold performance with equipment that's already past its design life.

Multi-site visibility. Each plant is a separate firefight. No way to compare performance, share learnings, or manage chemistry programs at scale.

We're CarboNet

A materials science company
focused on water treatment
chemistry.

Our customers are industrial brands and operators
who rely on, work with, and extract value from water.

Markets served :

- Construction
- Food & beverage
- Manufacturing
- Mining
- Municipal
- Oil & gas
- Pulp & paper

What we deliver:

Water treatment

Resource recovery



**Thankfully there are
NanoNets.**

The NanoNet platform produces water treatment products that require less equipment, chemicals, and labour. Operators can use them to radically reduce CAPEX, OPEX, and emissions.



Thankfully there are NanoNets.

The NanoNet platform produces water treatment products that require less equipment, chemicals, and labour. Operators can use them to radically reduce CAPEX, OPEX, and emissions.

SimpleFloc

Our first NanoNet product, a flocculant called SimpleFloc, removes make-down. Make-down is a messy, expensive ritual that's CAPEX and OPEX heavy. Removing it is like moving from gas to electric cars. It changes everything:

- Make-down chemicals are slashed by 90%
- Cost-to-treat is cut by 50%
- Crew hours can be cut 35-50%
- CAPEX savings run in the million
- Emissions are dramatically reduced
- Regulatory exposure is equally reduced

SimpleFloc removes the first and biggest constraint: make-down.

A pre-activated gel flocculant. No mixing system. No equipment bottleneck. Pumps directly into the line.

Make-down is a messy, expensive ritual—CAPEX and OPEX heavy, prone to failure, and a hard ceiling on throughput. Most flocculants can't function without it.

SimpleFloc is pre-activated. It arrives ready to pump, requires no dilution equipment, and removes the infrastructure constraint entirely. It changes the economics of every project:



Make-down chemicals cut by 70–90%



Cost-to-treat reduced by up to 50%



Crew hours cut 35–50%



CAPEX requirements substantially reduced



Permit compliance risk reduced alongside chemical load

SimpleFloc is the first in a suite built for the plant as it actually operates.

NanoNet

Platform chemistry that delivers stronger floc and faster settling at lower polymer volumes—even as conditions shift. The engine behind SimpleFloc's performance across variable influent.

SimplePrime

Coagulants that stabilize the front end of treatment—improving separation speed, reducing sludge volume, and smoothing pH variability before the flocculant ever reaches the line.

SpecialOps

A monitoring dashboard that gives multi-site operations a single view across facilities: dosing programs, chemistry performance, and site conditions—without dispatching a vehicle to every plant.

NanoNet. SimplePrime. SpecialOps.

Thankfully there are
NanoNets.

The NanoNet platform produces water treatment products that require less equipment, chemicals, and labour. Operators can use them to radically reduce CAPEX, OPEX, and emissions.

Select

Our second NanoNet product, a resource-recovery agent, is being applied to coal ponds. Pond remediation is a new EPA rule that creates enormous liability for operators. But Select can sequester the fly ash from ponds so it can be sold to make Portland cement, turning the liability into a \$40b opportunity.

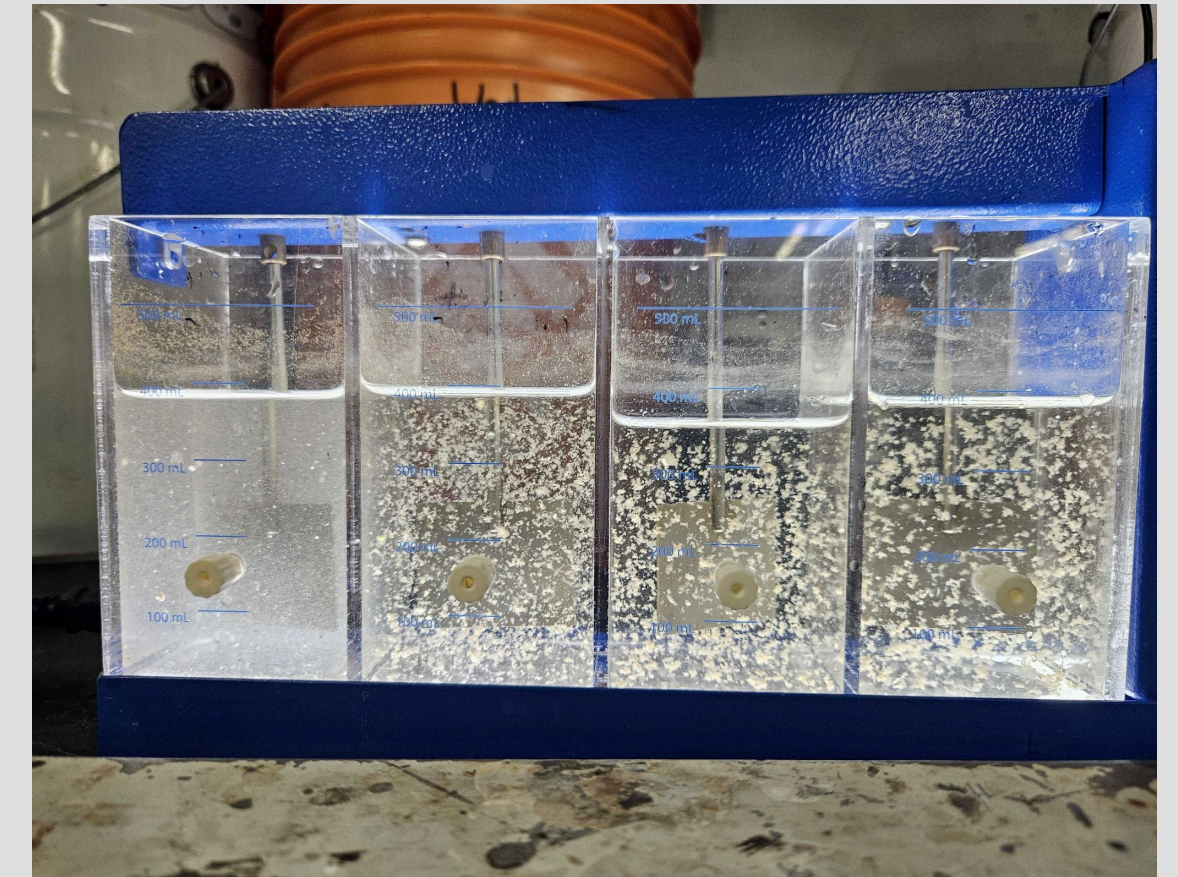


Thankfully there are
NanoNets.

The NanoNet platform produces water treatment products that require less equipment, chemicals, and labour. Operators can use them to radically reduce CAPEX, OPEX, and emissions.

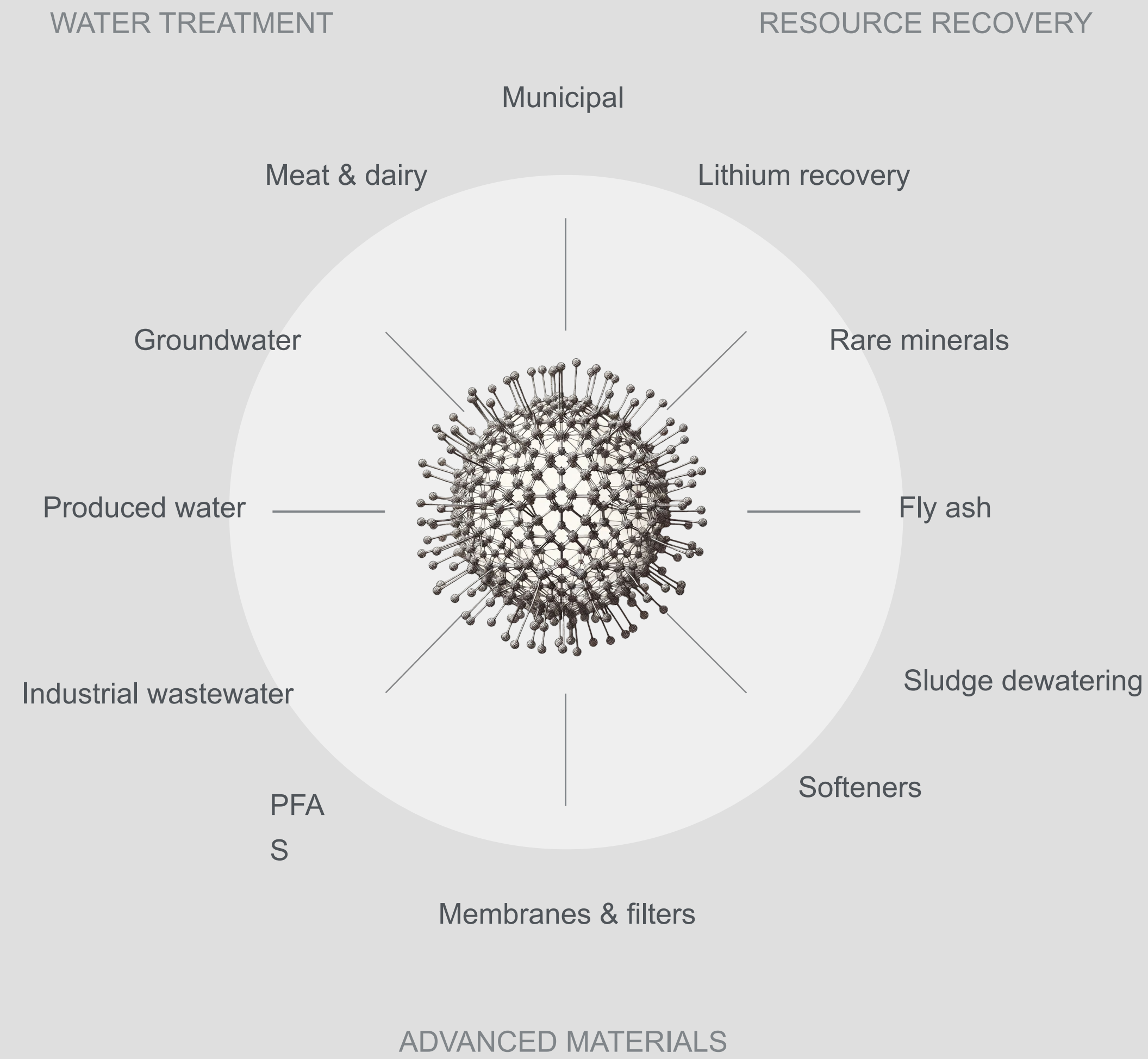
Bionic

Our most recent product, a 100% biodegradable and renewable polymer, is in municipal trial. Bio-based flocculants are the holy grail of water treatment. Replacing this old chemistry is a \$6b opportunity (7% CAGR)—and eliminates 2.5b lbs of petrochemicals annually.



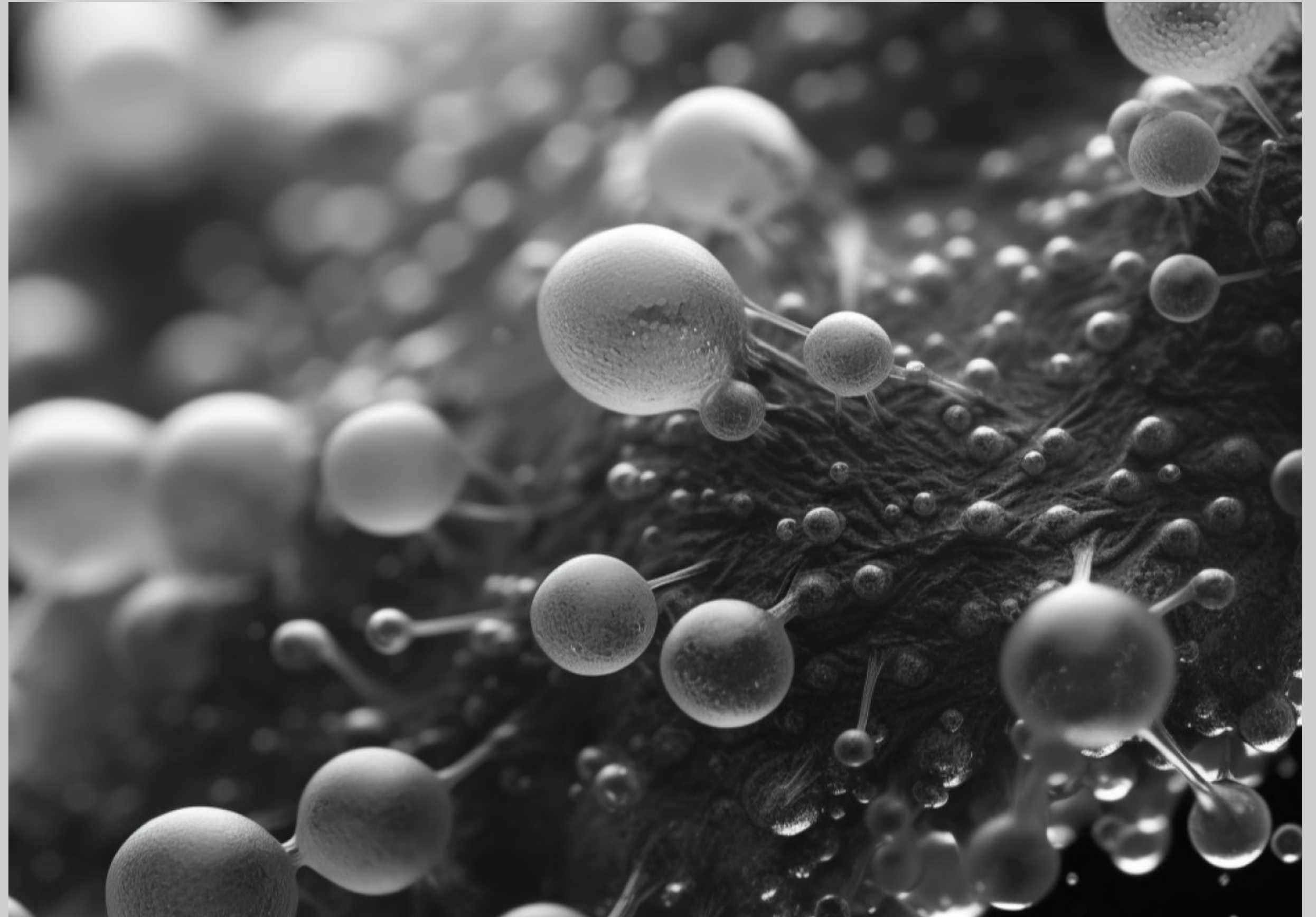
NanoNets have unlimited reach.

Increasingly powerful AI models allow us to rapidly sequence the optimal molecules for any liquid/solid separation challenge.



SimpleFloc helps municipal operators hit targets that incumbent chemistry couldn't reach.

Here are two case studies.



A municipal operator needed to dredge and dewater a lagoon but didn't have enough make-down rigs to take the job using conventional chemistry. Without make-down, incumbent flocculants couldn't be applied. The job would have been declined.

SimpleFloc required no make-down. It was deployed directly from the drum, plugged into the existing line, and the project proceeded without the equipment constraint that would have otherwise blocked it.

- PAM use reduced by 75%
- Crew efficiency improved significantly with simplified handling
- No permits were breached throughout the project
- Opened a new category of work the operator can now compete for



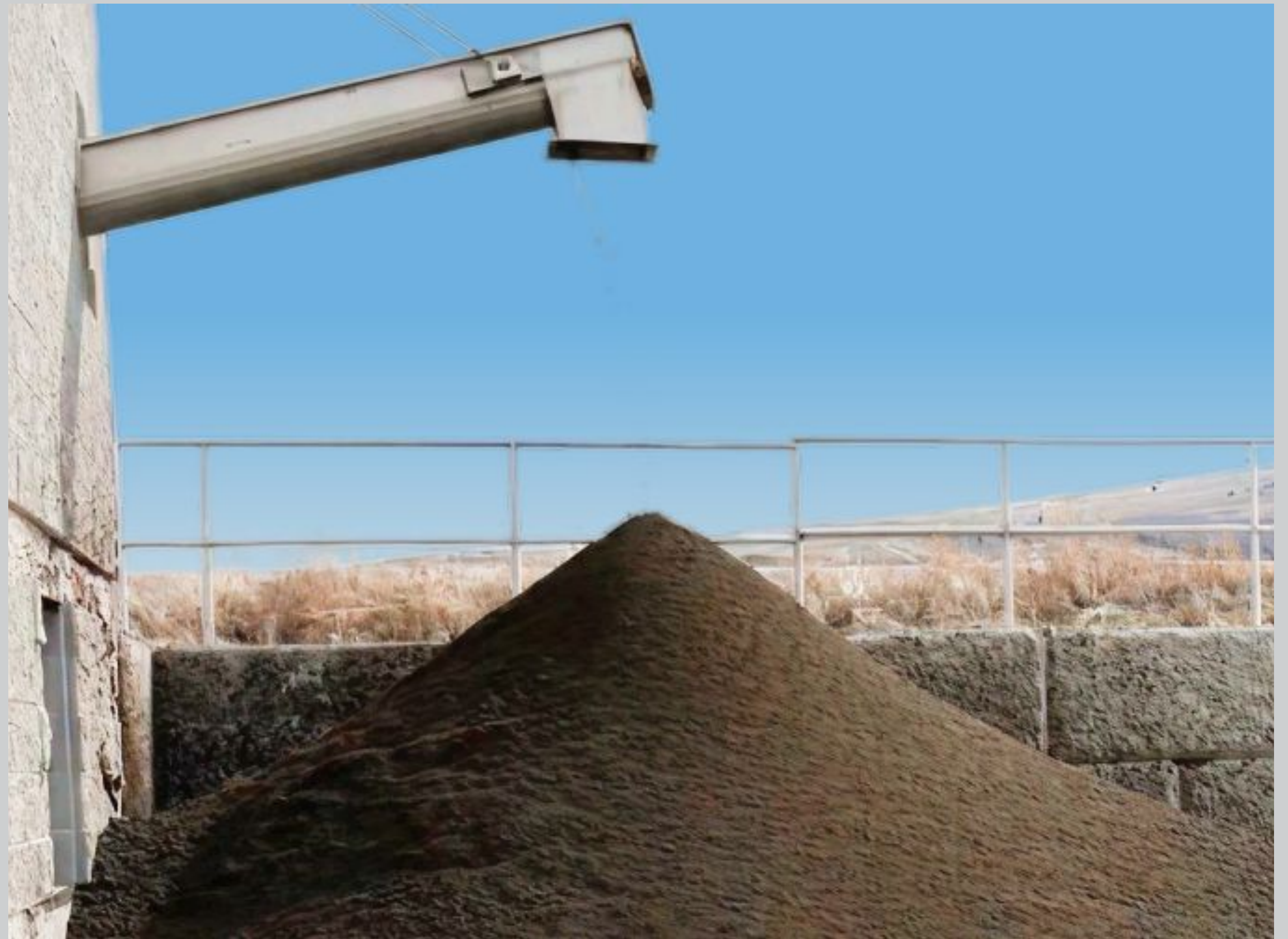
Regional Municipal Lagoon — Lagoon Dredging
CASE STUDY 1 / 2

Operators knew their new screw presses were capable of 17% cake solids dryness. Incumbent chemistry couldn't deliver it—regardless of dosing. After switching to SimpleFloc—combined with process improvements CarboNet's team identified—cake solids dryness reached 16–18% consistently.

- 6–8% increase in cake solids dryness
- 33% reduction in transportation cost
- 15% reduction in active polymer used
- Improved sludge stackability reduced on-site inventory pressure during farm application blackout periods

"CarboNet's team took the time to understand our process and really dial things in. Great customer service."

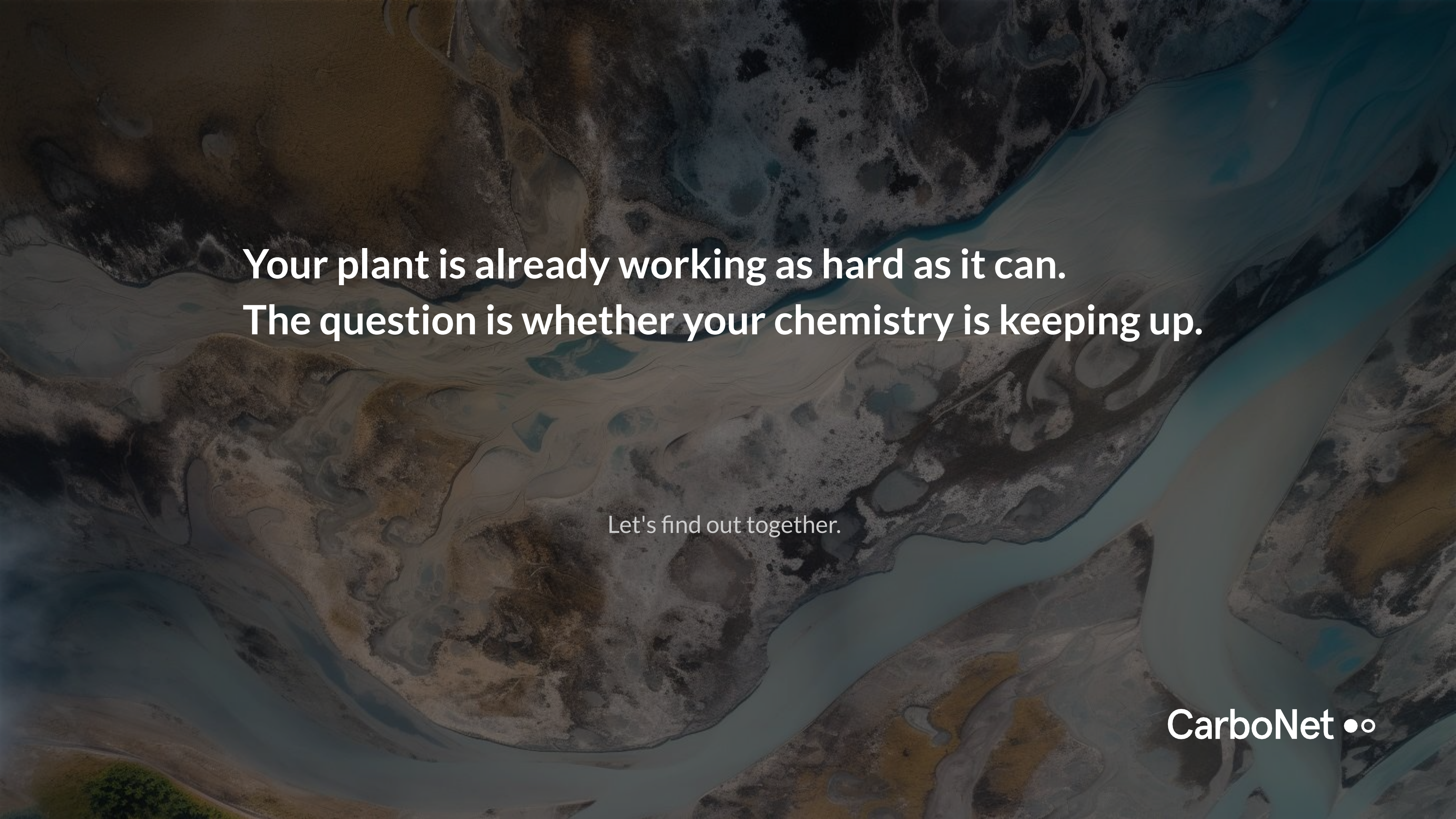
— Plant Operator, Idaho WPCF



“The most technologically innovative water treatment application I have laid eyes on this decade.”

Randy Khalil, Dewatering & Water Treatment SME





**Your plant is already working as hard as it can.
The question is whether your chemistry is keeping up.**

Let's find out together.

CarboNet ●●

What People are Saying about CarboNet

**“The most technologically innovative water treatment application
I’ve laid my eyes on”**

Construction dewatering SME

**“Our objective is to move CarboNet from 2% of our chemistry sales
to 60% in 2024”**

CEO / Sludge dewatering company

**“The results we achieved were outstanding. This saves us tens of
thousands in equipment costs – and the experience was seamless”**

Lead / Manufacturing market

“Magic. I’ve never seen anything like it in my career”

PM Lead / Materials Reclamation market

**“We’ve been trying, without success, for over 30 years to do
something like this”**

Product Manager / Global Chemicals Company

SimpleFloc helps municipal operators hit targets that incumbent chemistry couldn't reach.

Here are two examples.

1 / 2

Idaho Water Pollution Control Facility

Operators knew their new screw presses were capable of 17% cake solids dryness. Incumbent chemistry couldn't deliver it—regardless of dosing. After switching to SimpleFloc—combined with process improvements CarboNet's team identified—cake solids dryness reached 16–18% consistently.

- 6–8% increase in cake solids dryness
- 33% reduction in transportation cost
- 15% reduction in active polymer used
- Improved sludge stackability reduced on-site inventory pressure during farm application blackout periods

"CarboNet's team took the time to understand our process and really dial things in. Great customer service."

— Plant Operator, Idaho WPCF

SimpleFloc helps municipal operators hit targets that incumbent chemistry couldn't reach.

Here are two examples.

2 / 2

Regional Municipal Lagoon — Dredging & Dewatering

"It's the most technologically innovative water treatment application I have laid eyes on this decade. It's the difference between a system that just meets regulations and one that's set up to accelerate and scale."

— Randy Khalil, Dewatering & Water Treatment SME

ACME1 is a midstream operator competing in the Permian. They treat a significant share of the Basin's produced water and run equipment and crew across hundreds of sites.

SimpleFloc changed their P&L. They saved millions by removing make-down equipment, cutting crew by 35%, and eliminating 90% of polyacrylamide (PAM) from their chemical costs.

Cutting PAM meant big savings, but it also lowered the chance of breaching permits within the operating envelope.

Competition in O&G is fierce and margins are tight in produced water. SimpleFloc changed the delta on which XRI competes.



ACME3 is one of the world's largest engineering firms.

When the EPA recently told coal companies to remediate 8b tons of coal ponds, Aecom relied on NanoNets to create a targeting agent that would help separate fly ash from the muck.

Fly ash is a principal ingredient in Portland cement and 2b tons can be extracted from the ponds and sold to industry.

The value to Aecom is \$40b—an opportunity that wouldn't exist without NanoNets.



ACME4 delivers some 300,000 barrels of petroleum daily to the West Coast.

When they added 900km of new pipeline last year, NanoNets were crucial to ground dewatering, slashing PAM by 80%, reducing Scope 3 CO2 emissions by 77%, and conserving 8.4m litres of water.

Removing make-down also improved crew performance and eliminated the all-too-common equipment malfunctions that can cause delays, swap-outs, and shut-downs.

